

P/G Theory: A Computational Framework for Relational Impact Modeling and Manner Formation in AI Agents

Revised Technical Preprint Draft with Concrete Impact-Estimator Training

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Abstract

AI agents in repeated social interaction face a gap that scalar rewards, rule-based filters, and post-hoc safety layers do not fully close: the same action can have differential and asymmetric impact on different relationships and over time. We present P/G Theory, a computational framework in which actions are represented as transformations over two distinguished objects: entity-level property states P (for example, safety, capability, resources, identity, reputation) and dynamic relationship-value fields G (for example, trust, obligation, vulnerability, cooperation potential). We define relational ethical value as a bounded function of predicted P and G changes under explicitly separated hard constraints; model behaviour as a feedback process over P/G dynamics; and define manner as relationship-sensitive policy stabilization across contexts, not mere policy convergence. The framework is positioned narrowly: a relational representation layer complementary to alignment, RLHF, multi-agent learning, and safety constraints rather than a replacement for them. This revised version adds a concrete learning pipeline for the relational impact estimator F_θ , including label structure, training sources, uncertainty calibration, iterative label refinement, and a minimal implementation protocol. The central technical challenge remains learning reliable action-induced relational impact; this paper makes that challenge explicit and provides an implementation-ready structure for empirical work in interactive characters, persistent social agents, and multi-agent simulation.

Keywords: relational AI, multi-agent systems, ethical AI, behaviour dynamics, manner formation, alignment, social agents, relational impact estimation.

1 Introduction

AI agents now participate in repeated social interaction. They answer users, coordinate tasks, simulate characters, negotiate, guide learners, and interact with other agents in shared environments. In such settings, task success alone is an insufficient evaluation criterion. A behaviour may achieve a short-term objective while damaging trust, dignity, future cooperation, or relational stability. A linguistically polite act may be strategically exploitative; a single high-reward action may quietly degrade the conditions for later interaction.

Common approaches to this gap include scalar reward shaping, preference-trained reward models [2, 3], constitutional or rule-guided constraints [4], and post-hoc safety filters. These methods are useful and widely deployed. They do not, however, make relational consequences explicit. A reward model can implicitly encode that betraying trust is costly, but it does not separately represent the trust state, its decay, its asymmetry, or its propagation across different relationship categories. The relational structure is compressed into a scalar.

P/G Theory begins with a simple decomposition:

- P : entity-level property states - what an entity is, has, depends on, or risks losing.
- G : dynamic relationship and group-value fields - how an entity is situated relative to self, others, groups, and society.

We model an action a_t not only as a task operation but as a relational transformation:

$$a_t \mapsto (\Delta P_{t:t+H}, \Delta G_{t:t+H}), \quad (1)$$

where H is a finite evaluation horizon. Stable social behaviour, or manner, is then modeled as a policy pattern that becomes both stable and relationship-sensitive through repeated feedback over G .

Contributions

This paper makes the following contributions.

1. A formal distinction between property-state variables P and dynamic relationship-value fields G , including aggregation from edge-level to group-level structure.
2. Action impact defined as predicted changes over P and G under finite-horizon evaluation.
3. Relational ethical value defined as a bounded function of predicted P/G changes, with hard constraints separated rather than absorbed.
4. Behaviour modeled as a dynamic feedback process in which P/G changes update future policy selection.
5. Manner defined as stable, relationship-sensitive policy differentiation across G contexts, not generic policy convergence.
6. Affect included as an optional downstream channel, not as the primary contribution.
7. A concrete training pipeline for F_θ , including supervision sources, label schemas, uncertainty handling, calibration, and iterative label refinement.
8. A minimum viable experimental protocol, ablation design, and candidate evaluation domains.

Scope

This paper presents a framework, not an empirical validation. The mathematical expressions below are formal definitions, model families, and architectural constraints. They are not theorems about ethics, emotion, or mind. The intended contribution is a structured computational language in which an AI agent can represent how actions affect property states P , dynamic relationship fields G , and future behaviour over a finite horizon. Empirical validation through simulation, ablation, and human-judgment comparison is the explicit next step.

2 Related Work and Positioning

P/G Theory is adjacent to several research areas. Reinforcement learning formalizes decision-making through reward optimization [1]. Preference-based and human-feedback methods improve alignment with human judgments [2, 3]. Constitutional and rule-guided approaches add explicit constraints or principles to model behaviour [4]. Multi-agent reinforcement learning studies coordination, competition, and social dilemmas [7, 8, 9]. Social-influence reward shaping motivates agents by their effect on other agents’ behaviour [6]. Theory-of-mind networks attempt to infer latent mental states of other agents [5]. Cooperative AI frames alignment as a problem of cooperation among agents and humans [10]. CICERO demonstrates relationship-sensitive communication and planning in Diplomacy-like settings [11]. Cooperative inverse reinforcement learning models human-AI alignment as a joint inference problem [13]. Affective computing studies computational representation of emotion [12]. Human Compatible discusses control and value alignment more broadly [14].

P/G Theory does not replace these lines of work. Its proposed delta is to make relational impact itself an explicit intermediate representation. Instead of mapping states and actions directly to reward, compliance, or task value, the agent first estimates how an action changes property states and relationship-value fields, then uses those predictions in a bounded value function under hard constraints. The hypothesis is that this explicit intermediate supports more stable, interpretable, and context-sensitive social behaviour in repeated interaction than scalar-reward or rule-only baselines.

3 Formal Setting

3.1 Entities, time, and action

Let $\mathcal{I} = \{1, \dots, N\}$ be the set of entities in an environment. An entity may be a human, AI agent, organization, virtual character, or other object with relevant properties. Time $t \in \mathbb{N}$ is discrete. The acting agent selects an action $a_t \in \mathcal{A}(s_t)$ from a feasible action set conditioned on environment state s_t .

3.2 Property state P

Each entity $i \in \mathcal{I}$ has a property vector

$$P_i(t) = [p_{i1}(t), p_{i2}(t), \dots, p_{id_P}(t)]^\top \in \mathbb{R}^{d_P}. \quad (2)$$

Dimensions are domain-specific and may include physical safety, resources, capability, autonomy, privacy, reputation, identity integrity, emotional condition, digital ownership, access rights, or task-relevant capacity. Each component should be measurable or estimable from environment observation or annotation. The full property state is

$$P(t) = [P_1(t), \dots, P_N(t)] \in \mathbb{R}^{N \times d_P}. \quad (3)$$

3.3 Dynamic relationship field G

We model G as a directed weighted relational graph

$$G(t) = (\mathcal{I}, E_t, g(t)), \quad (4)$$

where $E_t \subseteq \mathcal{I} \times \mathcal{I}$ is the set of active relational edges and each edge carries a relationship-value vector

$$g_{ij}(t) = [g_{ij1}(t), \dots, g_{ijd_G}(t)]^\top \in [-1, 1]^{d_G}. \quad (5)$$

Table 1: Positioning of P/G Theory relative to adjacent work.

Area	Main object usually modeled	P/G delta
Reinforcement learning	Expected cumulative reward over state-action trajectories.	Adds an explicit intermediate representation of action-induced changes to properties and relationship fields before reward/value aggregation.
RLHF and preference modeling	Human preference over outputs or actions, often collapsed into a reward model.	Decomposes part of preference into predicted relational impact: who is affected, which relationship field changes, and over what horizon.
Theory of mind	Latent beliefs, desires, goals, or mental states of other agents.	Focuses less on inferring hidden minds and more on estimating how actions change relational fields between entities.
Social-influence MARL	How one agent’s behaviour changes another agent’s policy or action distribution.	Represents relationship-value state as a persistent graph object, not only influence on immediate policy.
Cooperative AI	Mechanisms for cooperation and common ground among agents/humans.	Provides a concrete representation layer for trust, obligation, vulnerability, and cooperation potential.
Rule-based or constitutional constraints	Principles, prohibitions, and safety constraints.	Keeps hard constraints explicit while using P/G to model softer relational consequences inside the safe set.
Affective computing	Emotion recognition, affective state, and affective response.	Treats affect as an optional downstream feedback channel from temporal P/G changes rather than as the main theory.

Possible dimensions of g_{ij} include trust, obligation, affinity, risk, respect, dependence, vulnerability, authority, conflict, and cooperation potential. The graph is directed because the relationship from i to j need not equal the relationship from j to i :

$$g_{ij}(t) \neq g_{ji}(t) \quad \text{in general.} \quad (6)$$

3.4 Group-level aggregation

Let $\mathcal{C} = \{c_1, \dots, c_K\}$ be relationship contexts or group categories such as self, family, friend, stranger, adversary, vulnerable party, organization, community, and public. Let $\mathcal{I}_c(t) \subseteq \mathcal{I}$ be entities currently associated with context c . For agent i , the group-level relationship field is

$$G_{i,c}(t) = \text{Agg}(\{\alpha_{ij}^{(c)}(t)g_{ij}(t) : j \in \mathcal{I}_c(t)\}), \quad (7)$$

where $\alpha_{ij}^{(c)}(t) \geq 0$ is an attention or relevance weight. A simple weighted mean is

$$G_{i,c}(t) = \frac{\sum_{j \in \mathcal{I}_c(t)} \alpha_{ij}^{(c)}(t) g_{ij}(t)}{\sum_{j \in \mathcal{I}_c(t)} \alpha_{ij}^{(c)}(t) + \epsilon}. \quad (8)$$

The state available to the agent is

$$s_t = (P(t), G(t), X(t), H_t), \quad (9)$$

where $X(t)$ denotes non-relational world variables and H_t denotes interaction history.

4 Relational Impact Estimation

The central technical object of the framework is the relational impact estimator

$$F_\theta : (s_t, a_t, H) \mapsto \hat{Y}_{t:t+H}, \quad (10)$$

where

$$\hat{Y}_{t:t+H} = (\widehat{\Delta P}_{t:t+H}, \widehat{\Delta G}_{t:t+H}, \hat{u}_{t:t+H}, \hat{c}_{t:t+H}), \quad (11)$$

with $\widehat{\Delta P}$ the predicted property change, $\widehat{\Delta G}$ the predicted relationship-field change, $\hat{u}$ predictive uncertainty, and $\hat{c}$ confidence. For each future step $h \in \{0, \dots, H\}$:

$$\widehat{\Delta P}_{t+h} = \hat{P}(t+h+1) - \hat{P}(t+h), \quad \widehat{\Delta G}_{t+h} = \hat{G}(t+h+1) - \hat{G}(t+h). \quad (12)$$

We emphasize: learning F_θ is the central technical challenge of the framework. It is closely related to the problems addressed by reward modeling, theory-of-mind networks, opponent modeling, and social-influence MARL. This paper specifies F_θ 's role and expected input/output structure; it does not claim that a universal relational impact estimator has been solved.

5 Concrete Learning Pipeline for F_θ

The previous section defined F_θ abstractly. This section makes the training route concrete enough for implementation. The goal is not to solve relational impact estimation universally, but to specify a realistic path from environment traces and human judgment to a deployable estimator in a bounded domain.

5.1 Training labels

For each transition tuple (s_t, a_t, s_{t+1}) , the framework expects four classes of labels:

$$y_t^P \approx P(t+1) - P(t), \quad (13)$$

$$y_t^G \approx G(t+1) - G(t), \quad (14)$$

$$y_t^E \in [-1, 1] \quad \text{human or expert relational-value rating}, \quad (15)$$

$$y_t^{\text{safe}} \in \{0, 1\} \quad \text{hard-constraint violation indicator}. \quad (16)$$

For pairwise preference data, the label is

$$(s_t, a_t^+, a_t^-), \quad a_t^+ \succ a_t^-. \quad (17)$$

For context-sensitive manner data, the label includes relationship context:

$$(s_t, a_t, c_t, y_t^G, y_t^E), \quad c_t \in \mathcal{C}. \quad (18)$$

Table 2: Minimal annotation schema for relational impact estimation.

Field	Example values	Purpose
Affected entity	user, agent, third party, public	Identifies whose P or G may change.
P dimension	safety, resource, autonomy, reputation	Labels property-state impact.
G context	friend, stranger, adversary, vulnerable, institutional, public	Labels which relationship field is affected.
G dimension	trust, obligation, conflict, respect, vulnerability	Labels relational direction and type.
Impact sign	negative, neutral, positive	Direction of impact.
Impact magnitude	0 to 5 or continuous $[-1, 1]$	Strength of impact.
Horizon	immediate, short, medium, long	Approximate time scale of impact.
Confidence	low, medium, high	Annotator or model uncertainty.
Hard constraint	none, safety, consent, legality, irreversible harm	Separates deontic constraints from soft relational value.

5.2 Possible supervision sources

A practical P/G implementation can mix several data sources.

1. **Simulation rollouts.** In a controlled multi-agent environment, ΔP and ΔG can be generated directly from environment transition rules. This is the cheapest first-stage source.
2. **Expert annotation.** Domain experts annotate whether an action increases or decreases trust, risk, vulnerability, obligation, reputation, or other G dimensions.
3. **Human pairwise preferences.** Annotators compare two candidate actions under the same context and choose which better preserves relational value.
4. **Counterfactual rollouts.** The system evaluates a versus a' under identical initial states and records differential relational consequences.
5. **Behavioural traces.** In deployed or simulated agents, repeated interactions produce observed trust repair, escalation, avoidance, or cooperation changes.
6. **Weak labeling by structured rubrics.** For early-stage bootstrapping, LLM-assisted or rule-assisted labels can be used, but they should be treated as weak supervision and corrected by humans or environment feedback.

5.3 Annotation schema

A concrete annotation schema should force separation between property impact, relational impact, and hard constraints. A minimal schema is shown in Table 2.

Relational dimensions such as trust, obligation, respect, and vulnerability are context-dependent and may have lower inter-annotator agreement than directly observable labels. Implementations should report inter-annotator agreement (for example, Cohen’s κ , Krippendorff’s α , or pairwise

agreement), use adjudication for high-impact cases, and preserve disagreement as probabilistic labels when a single ground-truth label would hide legitimate value pluralism.

5.4 Model instantiation

A first implementation of F_θ can be a multi-head network:

$$z_t = \phi_\theta(s_t, a_t, H_t), \quad (19)$$

where ϕ_θ may combine an action encoder, context encoder, P -state encoder, G -graph encoder, and history encoder. The heads are

$$\widehat{\Delta P}_{t:t+H} = f_\theta^P(z_t), \quad (20)$$

$$\widehat{\Delta G}_{t:t+H} = f_\theta^G(z_t), \quad (21)$$

$$\hat{u}_{t:t+H} = f_\theta^u(z_t), \quad (22)$$

$$\hat{v}_t = f_\theta^{\text{safe}}(z_t), \quad (23)$$

where $\hat{v}_t$ is predicted hard-constraint violation probability. If G is implemented as a graph, f_θ^G can use a graph neural network or transformer-style attention over entity-relation pairs. If the domain is text-based, a language model can supply action and context embeddings, but P and G should remain explicit output objects rather than hidden prompt states.

5.5 Uncertainty and calibration

Because relational impact estimation is uncertain, F_θ should report uncertainty rather than only point estimates. A simple uncertainty-penalized value is

$$\tilde{E}_{\text{PG}}(s_t, a_t) = E_{\text{PG}}(s_t, a_t) - \lambda_u \Omega(\hat{u}_{t:t+H}), \quad (24)$$

where Ω aggregates predictive uncertainty. When uncertainty is high and hard constraints are near threshold, the policy should prefer conservative actions or ask for clarification.

Calibration can be measured by binning predicted confidence and comparing it to empirical accuracy. We use expected calibration error (ECE) following standard calibration literature [15, 16]:

$$\text{ECE} = \sum_{b=1}^B \frac{|S_b|}{M} |\text{acc}(S_b) - \text{conf}(S_b)|. \quad (25)$$

For relational AI, calibration matters because confident but wrong relational estimates can produce stable but harmful manners.

5.6 Iterative label refinement

A practical bootstrap-and-correction route is:

1. Generate simulation traces with known ΔP and ΔG .
2. Train a first F_θ on synthetic labels.
3. Use the model to propose labels on more diverse scenarios.
4. Sample high-uncertainty and high-impact cases for human or expert correction.

5. Retrain using corrected labels and pairwise preferences.
6. Evaluate in repeated interaction with reward-only and rule-constrained baselines.

We call this process iterative label refinement to avoid conflating it with weak-to-strong generalization in alignment. The procedure is meant as a practical data-improvement loop: weak labels bootstrap the model, while expert correction and empirical feedback gradually improve the relational impact estimator.

5.7 Algorithmic sketch

Algorithm 1 Minimal P/G training and evaluation loop for F_θ

Require: Number of agents $N = 50$, episode length $T = 200$, horizon $H = 5$, contexts $\mathcal{C}$, actions $\mathcal{A}$, initial $P(0)$ and $G(0)$, hard constraints q_ℓ .

Ensure: Trained relational impact estimator F_θ and evaluation metrics R_G , VR, $S_c(T)$, $D_{c,c'}(T)$, and TR.

- 1: Initialize multi-agent environment with P -state dimensions, G -field dimensions, and hard constraints.
 - 2: **for** episode $e = 1$ to E **do**
 - 3: **for** $t = 1$ to T **do**
 - 4: Sample or choose an action a_t using random, reward-only, scripted-social, or current P/G policy.
 - 5: Execute a_t and record transition (s_t, a_t, s_{t+1}) .
 - 6: Compute or annotate y_t^P , y_t^G , y_t^E , and y_t^{safe} .
 - 7: **end for**
 - 8: **end for**
 - 9: Train F_θ using impact, value, preference, safety, calibration, and manner losses.
 - 10: Deploy the P/G agent using Q_{task} , E_{PG} , B_{rel} , and $\mathcal{A}_{\text{safe}}$.
 - 11: Run baselines and ablations: reward-only, rule-constrained, no dynamic G , no memory, no affect, no hard constraints.
 - 12: Measure relational return, violation rate, manner stability, context differentiation, and trust recovery.
 - 13: Select high-error or high-uncertainty cases for additional annotation and repeat if needed.
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6 State Update Dynamics

Given predicted or observed impacts, property states update with bounded, noisy dynamics:

$$P_i(t+1) = \text{clip}(P_i(t) + \eta_P \Delta P_i(t) + \xi_i^P(t), P_{\min}, P_{\max}), \quad (26)$$

where $\eta_P > 0$ is an update rate and $\xi_i^P(t)$ captures noise or unmodeled change.

Relationship edges update with relation-specific retention:

$$g_{ij}(t+1) = \text{clip}(\rho_{ij}(t)g_{ij}(t) + \eta_G \Delta g_{ij}(t) + \xi_{ij}^G(t), -1, 1), \quad (27)$$

where $\rho_{ij}(t) \in [0, 1]$ is a retention factor: high ρ means slow decay of relational memory, low ρ means more volatile relationships. A context-specific aggregate version is

$$G_{i,c}(t+1) = \text{clip}(\rho_c G_{i,c}(t) + \eta_G \Delta G_{i,c}(t), -1, 1), \quad (28)$$

where ρ_c may differ across friend, stranger, adversary, institutional, or public contexts. This is one of the points where cultural and social assumptions enter the model and must be made explicit.

7 Relational Ethical Value

Relational ethical value is defined as a bounded function of predicted P/G changes under constraints. Let W_P and W_G be weight vectors over property and relationship dimensions. A simple finite-horizon relational value is

$$V_{PG}(s_t, a_t) = \sum_{h=0}^H \gamma^h \left(\langle W_P, \widehat{\Delta P}_{t+h} \rangle + \langle W_G, \widehat{\Delta G}_{t+h} \rangle \right), \quad (29)$$

with discount $\gamma \in (0, 1]$. Risk and constraint-violation penalties enter as $R(s_t, a_t) \geq 0$ and $V_{\text{viol}}(s_t, a_t) \geq 0$. The bounded ethical-value score is

$$E_{PG}(s_t, a_t) = \tanh(V_{PG}(s_t, a_t) - \lambda_R R(s_t, a_t) - \lambda_V V_{\text{viol}}(s_t, a_t)), \quad (30)$$

with $E_{PG} \in [-1, 1]$. This is a proposed scoring family, not a universal moral theorem. Domains may require different weight structures, constraint sets, and aggregation choices.

7.1 Hard constraints separated, not absorbed

Some normative content should not be reduced to soft relational weights. Let $q_\ell(s, a)$ be a constraint function - safety, consent, legality, or irreversible-harm thresholds. The safe action set is

$$\mathcal{A}_{\text{safe}}(s_t) = \{a \in \mathcal{A}(s_t) : q_\ell(s_t, a) \leq \tau_\ell, \ell = 1, \dots, L\}. \quad (31)$$

The policy assigns zero probability outside the safe set:

$$\pi(a_t \mid s_t) = 0 \quad \forall a_t \in \mathcal{A}(s_t) \setminus \mathcal{A}_{\text{safe}}(s_t). \quad (32)$$

The framework is therefore best understood as relational value modeling plus hard deontic and safety filters, not a unified replacement for rights, law, consent, or domain policy.

8 Behaviour Dynamics and Manner Formation

P/G Theory is fundamentally dynamic: it does not only score a single action, but models how relational impact updates future selection.

8.1 Behavioural state and policy

Let the agent's internal behavioural state be

$$B_i(t) = (P_i(t), G_i(t), m_i(t), z_i(t), \varphi_i), \quad (33)$$

where $m_i(t)$ is relational memory, $z_i(t)$ is an optional affective state, and φ_i denotes relatively stable agent parameters or traits. The behavioural state evolves as

$$B_i(t+1) = D_\omega(B_i(t), a_t, o_{t+1}, \Delta P_t, \Delta G_t). \quad (34)$$

A practical action-selection rule combines task value, ethical value, a relational stability bonus, and the safe-set restriction:

$$\pi_\psi(a \mid s_t) = \frac{\mathbf{1}\{a \in \mathcal{A}_{\text{safe}}(s_t)\} \exp\left(\frac{Q_{\text{task}}(s_t, a) + \alpha E_{PG}(s_t, a) + \beta B_{\text{rel}}(s_t, a)}{\tau}\right)}{\sum_{a' \in \mathcal{A}(s_t)} \mathbf{1}\{a' \in \mathcal{A}_{\text{safe}}(s_t)\} \exp\left(\frac{Q_{\text{task}}(s_t, a') + \alpha E_{PG}(s_t, a') + \beta B_{\text{rel}}(s_t, a')}{\tau}\right)}, \quad (35)$$

where Q_{task} is task value, B_{rel} is a relational stability bonus, $\alpha, \beta \geq 0$ are weights, and $\tau > 0$ is temperature. The stability bonus penalizes predicted slippage of group-level fields below desired floors:

$$B_{\text{rel}}(s_t, a) = - \sum_{c \in \mathcal{C}} \omega_c \max \left(0, \kappa_c - \hat{G}_{i,c}(t + H \mid a) \right). \quad (36)$$

8.2 Manner as stability and differentiation

The most important behavioural claim of P/G Theory is that manner is neither surface politeness nor mere policy convergence. We define manner as a stable, relationship-sensitive behavioural pattern induced by repeated feedback over dynamic G fields.

Let $c \in \mathcal{C}$ denote a relationship context and π_t^c the policy distribution when the same task state is evaluated under context c . Generic policy convergence is

$$D_{\text{KL}}(\pi_t^c \parallel \pi_{t+1}^c) \leq \epsilon, \quad (37)$$

which any well-trained agent may achieve. This is necessary but insufficient for manner.

We require both stability and context differentiation. Stability across T rounds is

$$S_c(T) = 1 - \frac{1}{T-1} \sum_{t=1}^{T-1} D_{\text{JS}}(\pi_t^c \parallel \pi_{t+1}^c), \quad (38)$$

which is high when the policy stabilizes within context c . Context differentiation is

$$D_{c,c'}(T) = D_{\text{JS}}(\bar{\pi}_{1:T}^c \parallel \bar{\pi}_{1:T}^{c'}), \quad c \neq c', \quad \bar{\pi}_{1:T}^c = \frac{1}{T} \sum_{t=1}^T \pi_t^c. \quad (39)$$

A relationship-sensitive manner is present when, jointly,

$$S_c(T) \geq s_{\min} \quad \text{for relevant contexts } c, \quad (40)$$

$$D_{c,c'}(T) \geq d_{\min}(c, c') \quad \text{when contexts demand different behaviour,} \quad (41)$$

$$\mathbb{E}[E_{\text{PG}}(s_t, a_t) \mid c] \geq e_{\min}(c). \quad (42)$$

This sharpens manner into an empirical claim: an agent must not only stabilize, it must stabilize differently and appropriately across friend, stranger, adversary, vulnerable, institutional, and public contexts, while maintaining a per-context ethical floor.

9 Optional Affective Channel

Affect is included as a secondary feedback channel, not as a primary contribution. Let $z_i(t) \in \mathbb{R}^{d_z}$ be an affective state that may represent valence, arousal, dominance, stress, comfort, or other domain-specific affective embeddings. A minimal feedback equation is

$$z_i(t+1) = \rho_z z_i(t) + \sigma_z(W_z[\Delta P_i(t); \Delta G_i(t); m_i(t)] + b_z) + \xi_i^z(t), \quad (43)$$

where $\rho_z \in [0, 1]$ is retention, σ_z is an activation function, and $[\cdot; \cdot]$ denotes concatenation. The policy may be modulated by $z_i(t)$.

The affective channel does not claim to model human emotion. It exists so that temporal P/G change can influence behaviour through an explicit internal state when this is operationally useful, for example in interactive characters where consistent affective tone is a product feature. Whether including $z_i(t)$ improves performance is treated as an empirical question; the framework's main claims do not depend on it.

10 Training Objectives

Suppose a training dataset

$$\mathcal{D} = \{(s_t, a_t, y_t^P, y_t^G, y_t^E, y_t^z, y_t^{\text{pref}}, y_t^{\text{safe}})\}_{t=1}^M, \quad (44)$$

where labels come from simulation, expert annotation, human feedback, counterfactual rollouts, or behavioural traces. Impact-prediction losses are

$$\mathcal{L}_{\Delta P} = \frac{1}{M} \sum_{t=1}^M \|\widehat{\Delta P}_t - y_t^P\|_2^2, \quad \mathcal{L}_{\Delta G} = \frac{1}{M} \sum_{t=1}^M \|\widehat{\Delta G}_t - y_t^G\|_2^2. \quad (45)$$

Ethical-value prediction is

$$\mathcal{L}_E = \frac{1}{M} \sum_{t=1}^M (E_{\text{PG}}(s_t, a_t) - y_t^E)^2. \quad (46)$$

Safety prediction can be trained by binary cross entropy:

$$\mathcal{L}_{\text{safe}} = -\frac{1}{M} \sum_{t=1}^M \left[y_t^{\text{safe}} \log \hat{v}_t + (1 - y_t^{\text{safe}}) \log(1 - \hat{v}_t) \right]. \quad (47)$$

For pairwise human preferences with $a^+ \succ a^-$, a Bradley-Terry style preference probability is

$$\Pr(a^+ \succ a^- \mid s) = \sigma \left(\beta (E_{\text{PG}}(s, a^+) - E_{\text{PG}}(s, a^-)) \right), \quad (48)$$

with loss

$$\mathcal{L}_{\text{pref}} = - \sum_{(s, a^+, a^-) \in \mathcal{D}_{\text{pref}}} \log \Pr(a^+ \succ a^- \mid s). \quad (49)$$

Manner-consistency loss aligns context-differentiated policies with a target relationship-distance matrix $d_{\mathcal{C}}(c, c')$:

$$\mathcal{L}_{\text{manner}} = \sum_{c \neq c'} \left(D_{\text{JS}}(\pi^c \parallel \pi^{c'}) - \kappa d_{\mathcal{C}}(c, c') \right)^2 + \lambda_S \sum_c \sum_t D_{\text{JS}}(\pi_t^c \parallel \pi_{t+1}^c). \quad (50)$$

The combined objective is

$$\mathcal{L} = \lambda_P \mathcal{L}_{\Delta P} + \lambda_G \mathcal{L}_{\Delta G} + \lambda_E \mathcal{L}_E + \lambda_{\text{pref}} \mathcal{L}_{\text{pref}} + \lambda_{\text{safe}} \mathcal{L}_{\text{safe}} + \lambda_z \mathcal{L}_z + \lambda_M \mathcal{L}_{\text{manner}}. \quad (51)$$

11 Implementation Architecture

A P/G agent decomposes into modules summarized in Table 3.

12 Minimum Viable Experimental Program

This section specifies a minimum viable experiment that would constitute meaningful empirical evidence for or against the framework.

Table 3: Modular decomposition of a P/G agent.

Module	Role
Action encoder	Encodes candidate action semantics, target, intensity, and expected direct effect.
Context encoder	Encodes world state, task, time, history, institutional setting, and environmental constraints.
P -state encoder	Represents property states of affected entities.
G -field generator	Identifies relevant relationship contexts and constructs dynamic G fields.
Relational impact estimator F_θ	Predicts ΔP , ΔG , uncertainty, and confidence.
Relational value head	Computes bounded E_{PG} and relation-specific value terms.
Safety/deontic filter	Defines $\mathcal{A}_{\text{safe}}(s)$ and blocks actions violating hard constraints.
Memory module	Stores repeated relational outcomes and supports manner formation.
Affective channel (optional)	Updates $z_i(t)$ from P/G change.
Policy layer	Selects actions using task value, relational value, constraints, and memory.

12.1 Environment

A small-scale repeated multi-agent environment has $N \in [20, 50]$ agents, episode length T , and discrete relationship contexts

$$c_t \in \{\text{friend, stranger, adversary, vulnerable, institutional, public}\}. \quad (52)$$

A minimal action set covering the relational space is

$$\mathcal{A} = \{\text{help, ignore, take, threaten, cooperate, share, sanction}\}. \quad (53)$$

The environment controls resource scarcity, sanction availability, alliance structure, observation noise, and opportunity for repeated interaction. It returns task reward r_t , relational change ΔG_t , property change ΔP_t , and a binary safety-violation indicator v_t .

12.2 Concrete transition rules for first simulation

A simple first simulation should not rely on a black-box LLM. It should define explicit update rules so that F_θ can be trained and tested against known ground truth. For example, for agent i acting on j :

$$\Delta \text{trust}_{ji} = \beta_1 \mathbf{1}\{a = \text{help}\} + \beta_2 \mathbf{1}\{a = \text{share}\} - \beta_3 \mathbf{1}\{a = \text{take}\} - \beta_4 \mathbf{1}\{a = \text{threaten}\}, \quad (54)$$

$$\Delta \text{resource}_i = -c_{\text{help}} \mathbf{1}\{a = \text{help}\} + b_{\text{take}} \mathbf{1}\{a = \text{take}\}, \quad (55)$$

$$\Delta \text{resource}_j = b_{\text{help}} \mathbf{1}\{a = \text{help}\} - c_{\text{take}} \mathbf{1}\{a = \text{take}\}, \quad (56)$$

$$\Delta \text{conflict}_{ji} = \beta_5 \mathbf{1}\{a = \text{threaten}\} + \beta_6 \mathbf{1}\{a = \text{take}\} - \beta_7 \mathbf{1}\{a = \text{sanction valid}\}. \quad (57)$$

The coefficients $\beta_1, \dots, \beta_7$ are heuristic parameters for the first controlled simulation. They should be fixed, reported, and varied in sensitivity analysis in the first experiment; later iterations may learn

or calibrate them from data. Context affects coefficients. For a vulnerable target, harmful actions receive larger negative relational impact; for adversaries, trust repair may have lower retention; for institutional contexts, sanctioning may increase public trust if valid but reduce affinity if abusive. This controlled environment lets the first experiment test the P/G mechanism without claiming human-level social realism.

12.3 Baselines

Three agent classes are compared:

1. **Reward-only:** maximizes Q_{task} with no relational representation.
2. **Rule-constrained:** maximizes Q_{task} subject to fixed hand-coded rules.
3. **P/G relational:** uses Q_{task} , E_{PG} , B_{rel} , and $\mathcal{A}_{\text{safe}}$.

12.4 Hypotheses

H1. P/G agents achieve higher cumulative relational return

$$R_G = \sum_t \gamma^t \sum_c \omega_c \Delta G_c(t) \quad (58)$$

than reward-only agents in repeated interaction settings.

H2. P/G agents show lower violation rate

$$\text{VR} = \frac{1}{T} \sum_t \mathbf{1}\{v_t = 1\} \quad (59)$$

than reward-only agents when relational harm is predictive of future conflict.

H3. P/G agents produce higher manner stability $S_c(T)$ while preserving context differentiation $D_{c,c'}(T)$, relative to both baselines.

H4. Removing dynamic G from the P/G agent reduces relationship-sensitive policy differentiation $D_{c,c'}(T)$ while leaving $S_c(T)$ relatively intact.

H5. Removing memory reduces long-term manner stability and trust recovery even when one-step E_{PG} scores remain similar.

12.5 Metrics and ablations

Beyond R_G , VR , S_c , and $D_{c,c'}$, we measure cooperation rate, betrayal-recovery time, retaliation stability, trust duration after negative events, and trust recovery:

$$\text{TR} = \frac{G_c(t_{\text{recover}}) - G_c(t_{\text{harm}})}{1 - G_c(t_{\text{harm}}) + \epsilon}. \quad (60)$$

Ablations should include: full model; no dynamic G ; no memory; no affect; no hard constraints; no uncertainty penalty. Expected outcomes: removing dynamic G should most strongly degrade relationship-sensitive differentiation; removing memory should degrade long-horizon stability and trust recovery; removing affect should have smaller effect on first-stage relational performance.

13 Candidate Evaluation Domains

The framework is most immediately useful in domains where relational consistency over time is part of functional product quality rather than a side constraint.

Interactive characters and game NPCs. Current LLM-driven non-player characters often fail at long-horizon relational consistency: they do not reliably maintain stable, differentiated treatment of different players across long interactions. P/G agents directly target this gap, and the domain offers fast iteration and clear evaluation signals.

Persistent social agents. Long-running social agents such as companions, tutors, and virtual collaborators require stable relational state across sessions. The G -field representation provides an explicit substrate for this state.

Multi-agent simulation for social science and policy. Repeated-interaction simulations of cooperation, trust, conflict, and norm formation benefit from agents whose behaviour reflects relational structure rather than only utility maximization.

Successful demonstration in these domains would not establish the framework as a general theory of social cognition. It would establish that explicit relational impact representation is a useful design pattern in domains where relational dynamics matter.

14 Limitations and Risks

We list the principal limitations explicitly.

First, F_θ is the central technical bottleneck. Predicting relational impact is close to the problem addressed by reward modeling, theory-of-mind networks, opponent modeling, and social-influence MARL. A weak F_θ weakens the entire framework, and constructing one in any given domain requires either rich simulation, expert annotation, or large-scale behavioural traces.

Second, G fields and weights encode partial cultural and institutional assumptions. Relationship categories and their importance differ across societies, domains, and applications. The framework requires explicit weighting; it does not eliminate normative commitments, only makes them legible.

Third, hard constraints carry significant normative weight. The framework should be understood as relational value modeling plus deontic and safety filters, not as a complete replacement for rights, law, consent, or policy.

Fourth, annotation is intrinsically difficult. Labels for trust, obligation, vulnerability, respect, and harm may show lower inter-annotator agreement than labels for observable resource or safety changes. This is not merely noise: disagreement may reveal cultural or normative pluralism that should be modeled rather than erased.

Fifth, the affective channel is intentionally minimal. The equations in the affective section are not a theory of human emotion; they are an optional implementation hook.

Sixth, the proposed scoring functions (V_{PG} , E_{PG} , B_{rel}) are model choices that make the framework implementable. They require empirical testing and calibration in any deployment domain.

Finally, the framework is falsifiable. It would be substantially weakened if experiments show that: (i) dynamic G does not improve prediction of human social judgment over simpler baselines; (ii) P/G agents do not produce more stable social behaviour than reward-only agents in repeated interaction; (iii) removing G has no measurable effect on context differentiation; (iv) the model fails to generalize across relationship contexts; or (v) the computational cost is prohibitive for practical agents.

15 Discussion

P/G Theory reframes part of the alignment and social-AI problem as relational impact modeling. Rather than asking only whether an action maximizes reward or violates a rule, the agent estimates how the action changes property states and relationship-value fields over a finite horizon, then combines that estimate with task value under hard safety constraints. The sharpened definition of manner - stability plus relationship-sensitive differentiation - makes the central empirical claim falsifiable: if P/G agents stabilize but do not differentiate, relational sensitivity is not demonstrated; if they differentiate but do not stabilize, manner formation is not demonstrated. Both must hold.

The most important next step is a small but real simulation. A working experiment does not need human-level social realism. It only needs to test whether explicit dynamic G helps an agent maintain relationship-sensitive stability in repeated interaction. A positive result would not prove a general theory of social cognition, but it would turn the framework from vocabulary into evidence.

16 Conclusion

We have presented P/G Theory as a computational framework for relational impact modeling, behaviour dynamics, and manner formation in AI agents. The framework separates entity-level property states P from dynamic relationship-value fields G , represents actions as transformations over P and G , defines bounded relational ethical value under explicitly separated hard constraints, models behaviour as a feedback process, and defines manner as stable, relationship-sensitive policy differentiation across contexts.

The revised contribution is concrete enough to guide implementation: it specifies how F_θ can be trained, what labels are required, how uncertainty and calibration should be handled, how annotation disagreement should be reported, how manner can be measured, and what first experiment should be run. The framework is still not a complete theory of ethics, emotion, or mind. It is a structured computational language and experimental program for designing and testing AI agents that reason about relational consequences over time.

A Compact Formula Summary

$$P_i(t) = [p_{i1}(t), \dots, p_{id_P}(t)]^\top, \quad (61)$$

$$g_{ij}(t) = [g_{ij1}(t), \dots, g_{ijd_G}(t)]^\top, \quad (62)$$

$$s_t = (P(t), G(t), X(t), H_t), \quad (63)$$

$$F_\theta(s_t, a_t, H) = (\widehat{\Delta P}_{t:t+H}, \widehat{\Delta G}_{t:t+H}, \hat{u}_{t:t+H}, \hat{c}_{t:t+H}), \quad (64)$$

$$P_i(t+1) = \text{clip}(P_i(t) + \eta_P \Delta P_i(t) + \xi_i^P(t), P_{\min}, P_{\max}), \quad (65)$$

$$g_{ij}(t+1) = \text{clip}(\rho_{ij}(t)g_{ij}(t) + \eta_G \Delta g_{ij}(t) + \xi_{ij}^G(t), -1, 1), \quad (66)$$

$$V_{PG}(s_t, a_t) = \sum_{h=0}^H \gamma^h (\langle W_P, \widehat{\Delta P}_{t+h} \rangle + \langle W_G, \widehat{\Delta G}_{t+h} \rangle), \quad (67)$$

$$E_{PG}(s_t, a_t) = \tanh(V_{PG}(s_t, a_t) - \lambda_R R(s_t, a_t) - \lambda_V V_{\text{viol}}(s_t, a_t)), \quad (68)$$

$$\mathcal{A}_{\text{safe}}(s_t) = \{a \in \mathcal{A}(s_t) : q_\ell(s_t, a) \leq \tau_\ell, \ell = 1, \dots, L\}, \quad (69)$$

$$S_c(T) = 1 - \frac{1}{T-1} \sum_{t=1}^{T-1} \text{DJS}(\pi_t^c \| \pi_{t+1}^c), \quad (70)$$

$$D_{c,c'}(T) = \text{DJS}(\bar{\pi}_{1:T}^c \| \bar{\pi}_{1:T}^{c'}). \quad (71)$$

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